

IEEE Guide for Performing Arc-Flash Hazard Calculations

1. Overview

1.1 Scope

This guide provides models and an analytical process to enable calculation of the predicted incident thermal energy and the arc-flash boundary. The process covers the collection of field data if applicable, consideration of power system operating scenarios, and calculation parameters. Applications include electrical equipment and conductors for three-phase alternating current (ac) voltages from 208 V to 15 kV. Calculations for single-phase ac systems and direct current (dc) systems are not a part of this guide, but some guidance and references are provided for those applications. Recommendations for personal protective equipment (PPE) to mitigate arc-flash hazards are not included in this guide.

1.2 Purpose

The purpose of the guide is to enable qualified person(s) to analyze power systems for the purpose of calculating the incident energy to which employees could be exposed during operations and maintenance work. Contractors and facility owners can use this information to help provide appropriate protection for employees in accordance with the requirements of applicable electrical workplace safety standards.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 242™, IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (*IEEE Buff Book*™).^{2,3,4,5}

²IEEE publications are available from the Institute of Electrical and Electronics Engineers (<http://standards.ieee.org/>).

³The IEEE standards or products referred to in [Clause 2](#) are trademarks owned by the Institute of Electrical and Electronics Engineers, Incorporated.

⁴IEEE 3000 Standards Collection® (formerly known as the IEEE Color Books®) is the family of industrial and commercial power systems standards organized into “dot” standards that cover specific technical topics, which have been reorganized and, in some cases, updated from the content of the IEEE Color Books (<https://ieeexplore.ieee.org/browse/standards/collection/ieee/power-and-energy/3000StandardsCollection>).

⁵IEEE 3004 Standards: Protection and Coordination covers material from IEEE Std 242 (*IEEE Buff Book*) and IEEE Std 1015 (*IEEE Blue Book*).

IEEE Std 551™, IEEE Recommended Practice for Calculating AC Short-Circuit Currents in Industrial and Commercial Power Systems (*IEEE Violet Book*™).⁶

IEEE Std 1584.1™, IEEE Guide for the Specification of Scope and Deliverable Requirements for an Arc-Flash Hazard Calculation Study in Accordance with IEEE Std 1584™.

IEEE Std C37.010™, IEEE Application Guide for AC High-Voltage Circuit Breakers > 1000 Vac Rated on a Symmetrical Current Basis.

3. Definitions, acronyms, and abbreviations

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.⁷

3.1 Definitions

arc: A plasma cloud formed in a gap between two electrodes with sufficient potential difference.

arc current: *See:* **arcing fault current**.

arc duration: *See:* **clearing time**.

arc flash: An electric arc event with thermal energy dissipated as radiant, convective, and conductive heat.

NOTE—See [Annex E](#) for additional information.⁸

arc-flash boundary: A distance from a prospective arc source at which the incident energy is calculated to be 5.0 J/cm² (1.2 cal/cm²).

arc-flash hazard: A dangerous condition associated with an electric arc likely to cause possible injury.

arc-flash hazard calculation: The use of equations to compute the incident energy at a specific working distance and the arc-flash boundary.

arcing fault current: A fault current flowing through an electrical arc plasma. *Syn:* **arc current**.

available short-circuit current: At a given point in a circuit, the maximum current that the power system can deliver through a given circuit to any negligible-impedance short circuit applied at the given point, or at any other point that causes the highest current to flow through the given point. “Available short-circuit current” and “bolted fault current” are equivalent for a zero fault impedance.

bolted fault: A short-circuit condition that assumes zero impedance exists at the point of the fault.

circuit: A conductor or system of conductors through which an electric current flows.

clearing time: The total time between the beginning of a specified overcurrent and the final interruption of the circuit at rated voltage. *Syn:* **arc duration**.

⁶IEEE 3002 Standards: Power Systems Analysis covers material from IEEE Std 551 (*IEEE Violet Book*) and IEEE Std 399 (*IEEE Brown Book*).

⁷*IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>.

⁸Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

NOTE 1—In regard to fuses, it is the sum of the minimum melting time of a fuse plus tolerance and the arcing time. In regard to circuit breakers with integral trip units (usually rated less than 1000 V), it is the sum of the sensor time, plus opening time and the arcing time. For circuit breakers with separate relaying (usually rated greater than 1000 V), it is the sum of the minimum relay time, plus contact parting time and the arcing time. Sometimes referred to as total clearing time or interrupting time.

NOTE 2—Arc duration is the interval of time between the instant of the first initiation of the arc and the instant of final arc extinction. Arc duration is usually the same or directly related to the clearing time. See 6.9.1 for special circumstances where arc duration may be different than clearing time.

electrode configuration: The orientation and arrangement of the electrodes used in the testing performed for the model development.

NOTE 1—Electrodes were placed in open-air (“OA”) or enclosed (“Box”) configurations (with open front end). Electrodes were also oriented vertically and horizontally. Open-tipped and barrier-terminated electrode configurations were also used.

NOTE 2—Refer to Annex G. The following electrode configurations (test arrangements) are defined and listed according to their order of use within the incident energy model:

- **VCB:** Vertical conductors/electrodes inside a metal box/enclosure
- **VCBB:** Vertical conductors/electrodes terminated in an insulating barrier inside a metal box/enclosure
- **HCB:** Horizontal conductors/electrodes inside a metal box/enclosure
- **VOA:** Vertical conductors/electrodes in open air
- **HOA:** Horizontal conductors/electrodes in open air

fault current: A current that flows from one conductor to ground or to another conductor owing to an abnormal connection (including an arc) between the two conductors.

incident energy: The amount of thermal energy impressed on a surface, a certain distance from the source, generated during an electric arc event.

NOTE 1—The incident energy is calculated at the working distance. Incident energy increases as the distance from the potential arc source decreases, and the incident energy decreases as the distance increases. See: **working distance**.

NOTE 2—The units used to measure incident energy are joules per square centimeter (J/cm^2) or calories per square centimeter (cal/cm^2). See B.2.

nominal voltage: A numerical value of a circuit or system for designating its voltage class. (National Electrical Safety Code® (NESC®) (Accredited Standards Committee C2-2012) [B1]⁹)

qualified person: A person who performs arc-flash hazard calculations by using skills and knowledge related to the construction and operation of the electrical equipment and installations and has experience in power system studies and arc-flash hazard analysis.

voltage (nominal): See: **nominal voltage**.

working distance: The distance between the potential arc source and the face and chest of the worker performing the task.

NOTE—Parts of the body closer to the potential arc source other than the face and chest receive a greater incident energy. The arc source is usually energized parts within an equipment enclosure or exposed energized parts in open air.

⁹The numbers in brackets correspond to those of the bibliography in Annex A.

3.2 Acronyms and abbreviations

ac	alternating current
CF	correction factor
dc	direct current
E.C.	electrode configuration
HCB	horizontal conductors/electrodes inside a metal box/enclosure
HOA	horizontal conductors/electrodes in open air
LV	low voltage
MCC	motor control center
MV	medium voltage
OA	open air
PDU	power distribution unit
PPE	personal protective equipment
TCC	time current characteristic
UPS	uninterruptible power supplies
VCB	vertical conductors/electrodes inside a metal box/enclosure
VCBB	vertical conductors/electrodes terminated in an insulating barrier inside a metal box/ enclosure
VOA	vertical conductors/electrodes in open air

4. Model for incident energy calculations

4.1 General

An empirically derived model is provided for incident energy calculations. Development of this model is discussed in [Annex G](#). This annex provides more definitions and explains the derivations of the coefficients, variables, and terms used in the equations presented in [4.4](#) to [4.10](#). The equations in the model may be embedded in a spreadsheet or commercial software program, because it may be impractical to solve them by hand.

4.2 Range of model

The following empirically derived model, based upon statistical analysis and curve-fitting programs as well as an understanding of electrical arc physics, is applicable for systems with the following parameter range:

- Voltages in the range of 208 V to 15 000 V, three-phase (line-to-line)
Tests were performed in laboratory conditions using selected open-circuit voltages (V_{oc}). While the model utilizes V_{oc} , pre-fault voltage (system nominal voltage, utilization voltage, etc.) can be used for application of this model.
- Frequency of 50 Hz or 60 Hz
- Bolted fault current (rms symmetrical)
 - 208 V to 600 V: 500 A to 106 000 A
 - 601 V to 15 000 V: 200 A to 65 000 A
- Gaps between conductors
 - 208 V to 600 V: 6.35 mm to 76.2 mm (0.25 in to 3 in)
 - 601 V to 15 000 V: 19.05 mm to 254 mm (0.75 in to 10 in)

- Working distances greater than or equal to 305 mm (12 in) (see G.7.6 for details on the lower limit)
- Fault clearing time: No limit (see G.7.8 for more details)
- Enclosures tested (with open front end) as shown in the following table:

Open-circuit voltage (V)	Enclosure dimensions (H × W × D)	
	SI units (metric)	Imperial units
600	508 mm × 508 mm × 508 mm	20 in × 20 in × 20 in
2 700	660.4 mm × 660.4 mm × 660.4 mm	26 in × 26 in × 26 in
14 300	914.4 mm × 914.4 mm × 914.4 mm	36 in × 36 in × 36 in

- Enclosure dimension limits (established using the enclosures from the 2002 version of this guide)
 - Maximum height or width: 1244.6 mm (49 in)
 - Maximum opening area: 1.549 m² (2401 in²)
 - Minimum width: The width of the enclosure should be larger than four times the gap between conductors (electrodes).
- Electrode configurations (see the definition of *electrode configuration* in 3.1 and Figure G.1 through Figure G.5)

There are alternative calculation methods for system parameters that fall outside of the range of the model. However, no particular recommendation can be made because there are other application details such as bolted fault current levels, voltage, gap length, operating frequency, number of phases, types of faults, etc. The user is advised to properly research alternative calculation methods and their application viabilities.

4.3 Model application overview

The model for incident energy calculations has been divided into the following two parts depending on the system open-circuit voltage, V_{oc} :

- Model for $600 \text{ V} < V_{oc} \leq 15\,000 \text{ V}$
- Model for $208 \text{ V} \leq V_{oc} \leq 600 \text{ V}$

Sustainable arcs are possible but less likely in three-phase systems operating at 240 V nominal or less with an available short-circuit current less than 2000 A.

The model uses a two-step process in which intermediate values of average arc current, incident energy, and arc-flash boundary are interpolated to determine final values. Correction factors for enclosure (box) size and arc current variation are applied to adjust the results.

A summary of the steps required to apply the model is provided as follows:

- a) To determine the arcing current
 - 1) Determine the applicable equipment electrode configuration based on 6.6.
 - 2) If the system voltage is $600 \text{ V} < V_{oc} \leq 15\,000 \text{ V}$, use Equation (1) to find intermediate values at 600 V, 2700 V, and 14 300 V. Use Equation (16), Equation (17), Equation (18), and the guidance provided in 4.9 to find the final value of the arcing current.
 - 3) If the system voltage is $208 \text{ V} \leq V_{oc} \leq 600 \text{ V}$, use Equation (1) to find the intermediate value (600 V only) and Equation (25) to find the final value. Guidance for the determination of the final arcing current is provided in 4.10.

- b) Determine the arc duration or fault clearing time using the arcing current determined in step a). Guidance for determining the arc duration is provided in 6.9.
- c) To determine the incident energy
 - 1) Determine the enclosure size correction factor using the guidance provided in 4.8.4.
 - 2) If the system voltage is $600 \text{ V} < V_{oc} \leq 15\,000 \text{ V}$, use Equation (3), Equation (4), and Equation (5) to find intermediate values. Use Equation (19), Equation (20), Equation (21), and the guidance provided in 4.9 to find the final value of the incident energy.
 - 3) If the system voltage is $208 \text{ V} \leq V_{oc} \leq 600 \text{ V}$, use Equation (6). Guidance for determining the final incident energy is provided in 4.10.
 - 4) Additional considerations are provided in 6.10.
- d) To determine the arc-flash boundary
 - 1) Determine the enclosure size correction factor per 4.8.
 - 2) If the system voltage is $600 \text{ V} < V_{oc} \leq 15\,000 \text{ V}$, use Equation (7), Equation (8), and Equation (9) to find intermediate values. Use Equation (22), Equation (23), Equation (24), and the guidance provided in 4.9 to find the final value of the arc-flash boundary.
 - 3) If the system voltage is $208 \text{ V} \leq V_{oc} \leq 600 \text{ V}$, use Equation (10). Guidance for the determination of the final arc-flash boundary is provided in 4.10.
- e) Use the guidance provided in 4.5 to account for the arcing current variation. Repeat step b), step c), and step d) using the reduced arcing current. It is possible that the incident energy and arc-flash boundary results obtained using the reduced arcing current are different. The final incident energy or arc-flash boundary is the higher of the two calculated values.

A set of sample calculations for different voltage levels is provided in Annex D to help illustrate the calculation process for two different system-voltage levels (4160 V and 480 V).

The equations presented in this guide can be applied to other unit systems by using the proper conversion factors.

4.4 Intermediate average arcing currents

The intermediate average arcing currents can be determined using Equation (1) as follows and the coefficients provided in Table 1. The arcing currents are calculated at three different open-circuit voltage (V_{oc}).

$$I_{arc_Voc} = 10^{(k1 + k2 \lg I_{bf} + k3 \lg G)} (k4 I_{bf}^6 + k5 I_{bf}^5 + k6 I_{bf}^4 + k7 I_{bf}^3 + k8 I_{bf}^2 + k9 I_{bf} + k10) \quad (1)$$

where

I_{bf}	is the bolted fault current for three-phase faults (symmetrical rms) (kA)
I_{arc_600}	is the average rms arcing current at $V_{oc} = 600 \text{ V}$ (kA)
I_{arc_2700}	is the average rms arcing current at $V_{oc} = 2700 \text{ V}$ (kA)
I_{arc_14300}	is the average rms arcing current at $V_{oc} = 14\,300 \text{ V}$ (kA)
G	is the gap distance between electrodes (mm)
$k1$ to $k10$	are the coefficients provided in Table 1
$\lg$	is $\log_{10}$

Table 1—Coefficients for Equation (1)

	E.C. / V_{oc}	k1	k2	k3	k4	k5	k6	k7	k8	k9	k10
VCB	600 V	-0.04287	1.035	-0.083	0	0	-4.783E-09	1.962E-06	-0.000229	0.003141	1.092
	2 700 V	0.0065	1.001	-0.024	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729
	14 300 V	0.005795	1.015	-0.011	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729
VCBB	600 V	-0.017432	0.98	-0.05	0	0	-5.767E-09	2.524E-06	-0.00034	0.01187	1.013
	2 700 V	0.002823	0.995	-0.0125	0	-9.204E-11	2.901E-08	-3.262E-06	0.0001569	-0.004003	0.9825
	14 300 V	0.014827	1.01	-0.01	0	-9.204E-11	2.901E-08	-3.262E-06	0.0001569	-0.004003	0.9825
HCB	600 V	0.054922	0.988	-0.11	0	0	-5.382E-09	2.316E-06	-0.000302	0.0091	0.9725
	2 700 V	0.001011	1.003	-0.0249	0	0	4.859E-10	-1.814E-07	-9.128E-06	-0.0007	0.9881
	14 300 V	0.008693	0.999	-0.02	0	-5.043E-11	2.233E-08	-3.046E-06	0.0001116	-0.001145	0.9839
VOA	600 V	0.043785	1.04	-0.18	0	0	-4.783E-09	1.962E-06	-0.000229	0.003141	1.092
	2 700 V	-0.02395	1.006	-0.0188	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729
	14 300 V	0.005371	1.0102	-0.029	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729
HOA	600 V	0.111147	1.008	-0.24	0	0	-3.895E-09	1.641E-06	-0.000197	0.002615	1.1
	2 700 V	0.000435	1.006	-0.038	0	0	7.859E-10	-1.914E-07	-9.128E-06	-0.0007	0.9981
	14 300 V	0.000904	0.999	-0.02	0	0	7.859E-10	-1.914E-07	-9.128E-06	-0.0007	0.9981

4.5 Arcing current variation correction factor

Calculate a second set of arc duration, using the reduced arcing current $I_{\text{arc_min}}$ to determine if the arcing current variation has an effect on the operating time of protective devices and consequently incident energy. The arcing current variation applies for all system open-circuit voltages within the valid range of the model (208 V to 15 000 V), but it is expected to have the most impact between 208 V and 600 V.

To determine a lower bound of the average rms arcing current, use Equation (2) as follows and the coefficients provided in Table 2:

$$I_{\text{arc_min}} = I_{\text{arc}} \times (1 - 0.5 \times \text{Var}C_f) \quad (2)$$

$$\text{Var}C_f = k1V_{\text{oc}}^6 + k2V_{\text{oc}}^5 + k3V_{\text{oc}}^4 + k4V_{\text{oc}}^3 + k5V_{\text{oc}}^2 + k6V_{\text{oc}} + k7$$

where

- $\text{Var}C_f$ is the arcing current variation correction factor
- I_{arc} is the final or intermediate rms arcing current(s) (kA) (see note)
- $I_{\text{arc_min}}$ is a second rms arcing current reduced based on the variation correction factor (kA)
- V_{oc} is the open-circuit voltage between 0.208 kV and 15.0 kV
- $k1$ to $k7$ are the coefficients provided in Table 2

Table 2—Coefficients for Equation (2)

E.C.	$k1$	$k2$	$k3$	$k4$	$k5$	$k6$	$k7$
VCB	0	−0.0000014269	0.000083137	−0.0019382	0.022366	−0.12645	0.30226
VCBB	1.138e-06	−6.0287e-05	0.0012758	−0.013778	0.080217	−0.24066	0.33524
HCB	0	−3.097e-06	0.00016405	−0.0033609	0.033308	−0.16182	0.34627
VOA	9.5606E-07	−5.1543E-05	0.0011161	−0.01242	0.075125	−0.23584	0.33696
HOA	0	−3.1555e-06	0.0001682	−0.0034607	0.034124	−0.1599	0.34629

NOTE—The correction factor $(1 - (0.5 \times \text{Var}C_f))$ is applied as follows:

- 208 V $\leq V_{\text{oc}} \leq$ 600 V: To I_{arc} (final current only)
- 600 V $< V_{\text{oc}} \leq$ 15 000 V: To $I_{\text{arc_600}}$, $I_{\text{arc_2700}}$, and $I_{\text{arc_14300}}$ (intermediate average arcing currents). The final I_{arc} value inherits the correction factor.

The “0.5” coefficient indicates that variation is applied to the average arcing current to obtain a lower-bound value arcing current.

4.6 Intermediate incident energy (E)

Use Equation (3) to Equation (6) as follows and Table 3, Table 4, and Table 5 to determine the intermediate incident energy values:

$$E_{600} = \frac{12.552}{50} T \times 10^{\left(k1 + k2 \lg G + \frac{k3 I_{\text{arc}_600}}{k4 I_{\text{bf}}^7 + k5 I_{\text{bf}}^6 + k6 I_{\text{bf}}^5 + k7 I_{\text{bf}}^4 + k8 I_{\text{bf}}^3 + k9 I_{\text{bf}}^2 + k10 I_{\text{bf}}} + k11 \lg I_{\text{bf}} + k12 \lg D + k13 \lg I_{\text{arc}_600} + \lg \frac{1}{CF} \right)} \quad (3)$$

$$E_{2700} = \frac{12.552}{50} T \times 10^{\left(k1 + k2 \lg G + \frac{k3 I_{\text{arc}_2700}}{k4 I_{\text{bf}}^7 + k5 I_{\text{bf}}^6 + k6 I_{\text{bf}}^5 + k7 I_{\text{bf}}^4 + k8 I_{\text{bf}}^3 + k9 I_{\text{bf}}^2 + k10 I_{\text{bf}}} + k11 \lg I_{\text{bf}} + k12 \lg D + k13 \lg I_{\text{arc}_2700} + \lg \frac{1}{CF} \right)} \quad (4)$$

$$E_{14300} = \frac{12.552}{50} T \times 10^{\left(k1 + k2 \lg G + \frac{k3 I_{\text{arc}_14300}}{k4 I_{\text{bf}}^7 + k5 I_{\text{bf}}^6 + k6 I_{\text{bf}}^5 + k7 I_{\text{bf}}^4 + k8 I_{\text{bf}}^3 + k9 I_{\text{bf}}^2 + k10 I_{\text{bf}}} + k11 \lg I_{\text{bf}} + k12 \lg D + k13 \lg I_{\text{arc}_14300} + \lg \frac{1}{CF} \right)} \quad (5)$$

$$E_{\leq 600} = \frac{12.552}{50} T \times 10^{\left(k1 + k2 \lg G + \frac{k3 I_{\text{arc}_600}}{k4 I_{\text{bf}}^7 + k5 I_{\text{bf}}^6 + k6 I_{\text{bf}}^5 + k7 I_{\text{bf}}^4 + k8 I_{\text{bf}}^3 + k9 I_{\text{bf}}^2 + k10 I_{\text{bf}}} + k11 \lg I_{\text{bf}} + k12 \lg D + k13 \lg I_{\text{arc}} + \lg \frac{1}{CF} \right)} \quad (6)$$

where

E_{600}	is the incident energy at $V_{\text{oc}} = 600$ V (J/cm ²)
E_{2700}	is the incident energy at $V_{\text{oc}} = 2700$ V (J/cm ²)
E_{14300}	is the incident energy at $V_{\text{oc}} = 14\,300$ V (J/cm ²)
$E_{\leq 600}$	is the incident energy for $V_{\text{oc}} \leq 600$ V (J/cm ²)
T	is the arc duration (ms)
G	is the gap distance between conductors (electrodes) (mm)
I_{arc_600}	is the rms arcing current for 600 V (kA)
I_{arc_2700}	is the rms arcing current for 2700 V (kA)
I_{arc_14300}	is the rms arcing current for 14 300 V (kA)
I_{arc}	is rms arcing current for $V_{\text{oc}} \leq 600$ V [obtained using Equation (25)] (kA)
I_{bf}	is bolted fault current for three-phase faults (symmetrical rms) (kA)
D	is the distance between electrodes and calorimeters (working distance) (mm)
CF	is correction factor for enclosure size ($CF = 1$ for VOA and HOA configurations)
$\lg$	is $\log_{10}$
$k1$ to $k13$	are the coefficients provided in Table 3, Table 4, and Table 5. For Equation (3) use Table 3, for Equation (4) use Table 4, for Equation (5) use Table 5, and for Equation (6) use Table 3

Table 3—Coefficients for Equation (3), Equation (6), Equation (7), and Equation (10)

600 V	k1	k2	k3	k4	k5	k6	k7	k8	k9	k10	k11	k12	k13
VCB	0.753364	0.566	1.752636	0	0	-4.783E-09	0.000001962	-0.000229	0.003141	1.092	0	-1.598	0.957
VCBB	3.068459	0.26	-0.098107	0	0	-5.767E-09	0.000002524	-0.00034	0.01187	1.013	-0.06	-1.809	1.19
HCBB	4.073745	0.344	-0.370259	0	0	-5.382E-09	0.000002316	-0.000302	0.0091	0.9725	0	-2.03	1.036
VOA	0.679294	0.746	1.222636	0	0	-4.783E-09	0.000001962	-0.000229	0.003141	1.092	0	-1.598	0.997
HOA	3.470417	0.465	-0.261863	0	0	-3.895E-09	0.000001641	-0.000197	0.002615	1.1	0	-1.99	1.04

Table 4—Coefficients for Equation (4) and Equation (8)

2700 V	k1	k2	k3	k4	k5	k6	k7	k8	k9	k10	k11	k12	k13
VCB	2.40021	0.165	0.354202	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729	0	-1.569	0.9778
VCBB	3.870592	0.185	-0.736618	0	-9.204E-11	2.901E-08	-3.262E-06	0.0001569	-0.004003	0.9825	0	-1.742	1.09
HCBB	3.486391	0.177	-0.193101	0	0	4.859E-10	-1.814E-07	-9.128E-06	-0.0007	0.9881	0.027	-1.723	1.055
VOA	3.880724	0.105	-1.906033	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729	0	-1.515	1.115
HOA	3.616266	0.149	-0.761561	0	0	7.859E-10	-1.914E-07	-9.128E-06	-0.0007	0.9981	0	-1.639	1.078

Table 5—Coefficients for Equation (5) and Equation (9)

14 300 V	k1	k2	k3	k4	k5	k6	k7	k8	k9	k10	k11	k12	k13
VCB	3.825917	0.11	-0.999749	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729	0	-1.568	0.99
VCBB	3.644309	0.215	-0.585522	0	-9.204E-11	2.901E-08	-3.262E-06	0.0001569	-0.004003	0.9825	0	-1.677	1.06
HCBB	3.044516	0.125	0.245106	0	-5.043E-11	2.233E-08	-3.046E-06	0.000116	-0.001145	0.9839	0	-1.655	1.084
VOA	3.405454	0.12	-0.93245	-1.557E-12	4.556E-10	-4.186E-08	8.346E-07	5.482E-05	-0.003191	0.9729	0	-1.534	0.979
HOA	2.04049	0.177	1.005092	0	0	7.859E-10	-1.914E-07	-9.128E-06	-0.0007	0.9981	-0.05	-1.633	1.151

4.7 Intermediate arc-flash boundary (AFB)

Use Equation (7) to Equation (10) as follows and Table 3, Table 4, and Table 5 to determine the intermediate arc-flash boundary values:

$$AFB_{600} = 10^{\left(\frac{k1 + k2 \lg G + \frac{k3 I_{arc_600}}{k4 I_{bf}^7 + k5 I_{bf}^6 + k6 I_{bf}^5 + k7 I_{bf}^4 + k8 I_{bf}^3 + k9 I_{bf}^2 + k10 I_{bf}} + k11 \lg I_{bf} + k13 \lg I_{arc_600} + \lg \left(\frac{1}{CF} \right) - \lg \left(\frac{20}{T} \right)}{-k12} \right)} \quad (7)$$

$$AFB_{2700} = 10^{\left(\frac{k1 + k2 \lg G + \frac{k3 I_{arc_2700}}{k4 I_{bf}^7 + k5 I_{bf}^6 + k6 I_{bf}^5 + k7 I_{bf}^4 + k8 I_{bf}^3 + k9 I_{bf}^2 + k10 I_{bf}} + k11 \lg I_{bf} + k13 \lg I_{arc_2700} + \lg \left(\frac{1}{CF} \right) - \lg \left(\frac{20}{T} \right)}{-k12} \right)} \quad (8)$$

$$AFB_{14300} = 10^{\left(\frac{k1 + k2 \lg G + \frac{k3 I_{arc_14300}}{k4 I_{bf}^7 + k5 I_{bf}^6 + k6 I_{bf}^5 + k7 I_{bf}^4 + k8 I_{bf}^3 + k9 I_{bf}^2 + k10 I_{bf}} + k11 \lg I_{bf} + k13 \lg I_{arc_14300} + \lg \left(\frac{1}{CF} \right) - \lg \left(\frac{20}{T} \right)}{-k12} \right)} \quad (9)$$

$$AFB_{\leq 600} = 10^{\left(\frac{k1 + k2 \lg G + \frac{k3 I_{arc_600}}{k4 I_{bf}^7 + k5 I_{bf}^6 + k6 I_{bf}^5 + k7 I_{bf}^4 + k8 I_{bf}^3 + k9 I_{bf}^2 + k10 I_{bf}} + k11 \lg I_{bf} + k13 \lg I_{arc} + \lg \left(\frac{1}{CF} \right) - \lg \left(\frac{20}{T} \right)}{-k12} \right)} \quad (10)$$

where

- AFB_{600} is the arc-flash boundary for $V_{oc} = 600$ V (mm)
- AFB_{2700} is the arc-flash boundary for $V_{oc} = 2700$ V (mm)
- AFB_{14300} is the arc-flash boundary for $V_{oc} = 14\,300$ V (mm)
- $AFB_{\leq 600}$ is the arc-flash boundary for $V_{oc} \leq 600$ V (mm)
- G is the gap between electrodes (mm)
- I_{arc_600} is the rms arcing current for 600 V (kA)
- I_{arc_2700} is the rms arcing current for 2700 V (kA)
- I_{arc_14300} is the rms arcing current for 14 300 V (kA)
- I_{arc} is the rms arcing current for $V_{oc} \leq 600$ V [obtained using Equation (25)] (kA)
- I_{bf} is the bolted fault current for three-phase faults (symmetrical rms) (kA)
- CF is the correction factor for enclosure size ($CF = 1$ for VOA and HOA configurations)
- T is the arc duration (ms)
- $\lg$ is $\log_{10}$
- $k1$ to $k13$ are the coefficients provided in Table 3, Table 4, and Table 5. For Equation (7) use Table 3, for Equation (8) use Table 4, for Equation (9) use Table 5, and for Equation (10) use Table 3

4.8 Enclosure size correction factor

4.8.1 General

The VCB, VCBB, and HCB equations were normalized for a 508 mm × 508 mm × 508 mm (20 in × 20 in × 20 in) enclosure. This subclause provides instructions on how to adjust the incident energy for smaller and larger enclosures using the correction factor (*CF*) determined from Equation (14) and Equation (15). The method for adjusting the incident energy based on the enclosure size is as follows:

- A set of equivalent height and width values are determined based on the system voltage, electrode configuration, the enclosure height, and width. The depth is not considered unless the width and height are both less than 508 mm (20 in) and the system voltage is less than 600 V. The depth is used to classify the enclosure type as “Typical” or “Shallow” (see 4.8.2).
- The enclosure type, equivalent height, and width are used to determine an equivalent enclosure size parameter, which determines the value of the enclosure size correction factor, *CF*.
- Enclosures with opening areas larger than 1244.6 mm × 1244.6 mm (49 in × 49 in) may be encountered in actual equipment. The correction factor for 1244.6 mm × 1244.6 mm (49 in × 49 in) can be used for such. If either the width or height (or both) exceed 1244.6 mm (49 in), treat them as 1244.6 mm (49 in) for this model application.

4.8.2 Determination of enclosure type—Typical or shallow

The enclosure is “Shallow” when the following conditions are met:

- a) The system voltage is less than 600 V ac.
- b) Both the height and width are less than 508 mm (20 in).
- c) The enclosure depth is less than or equal to 203.2 mm (8 in).

If any of these conditions are not met, the enclosure is considered “Typical.”

4.8.3 Determination of equivalent height and width

Once the enclosure type has been classified, the equivalent height and width need to be determined by comparing their values against specific ranges for each of the three electrode configuration. For certain ranges, the equivalent height and width are determined using Equation (11) and Equation (12) as follows:

$$\text{Width}_1 = \left(660.4 + (\text{Width} - 660.4) \times \left(\frac{V_{oc} + A}{B} \right) \right) \times 25.4^{-1} \quad (11)$$

$$\text{Height}_1 = \left(660.4 + (\text{Height} - 660.4) \times \left(\frac{V_{oc} + A}{B} \right) \right) \times 25.4^{-1} \quad (12)$$

where

Height ₁	is the equivalent enclosure height
Width ₁	is the equivalent enclosure width
Width	is the actual enclosure width (mm)
Height	is the actual enclosure height (mm)
V_{oc}	is the open-circuit voltage (system voltage) (kV)
A	is a constant equal to 4 for VCB and 10 for VCBB and HCB
B	is a constant equal to 20 for VCB, 24 for VCBB, and 22 for HCB

Table 6 provides the guidelines to determine the equivalent enclosure height and width (Height₁ and Width₁) for different ranges of enclosure dimensions and electrode configurations.

Table 6—Guidelines to determine the equivalent height and width

E.C.	Range	<508 (mm)	≥508 and ≤660.4 (mm)	>660.4 and ≤1244.6 (mm)	>1244.6 (mm)
VCB	Width _i	= 20 (if Typical) = 0.03937 × Width (if Shallow ^a)	= 0.03937 × Width	obtained from Equation (11) and actual Width	obtained from Equation (11) with Width = 1244.6 mm
	Height _i	= 20 (if Typical) or = 0.03937 × Height (if Shallow ^a)	= 0.03937 × Height	= 0.03937 × Height	= 49
VCBB	Width _i	= 20 (if Typical) or = 0.03937 × Width (if Shallow ^a)	= 0.03937 × Width	obtained from Equation (11) and actual Width	obtained from Equation (11) with Width = 1244.6 mm
	Height _i	= 20 (if Typical) or = 0.03937 × Height (if Shallow ^a)	= 0.03937 × Height	obtained from Equation (12) and actual Height	obtained from Equation (12) with Height = 1244.6 mm
HCB	Width _i	= 20 (if Typical) or = 0.03937 × Width (if Shallow ^a)	= 0.03937 × Width	obtained from Equation (11) and actual Width	obtained from Equation (11) with Width = 1244.6 mm
	Height _i	= 20 (if Typical) or = 0.03937 × Height (if Shallow ^a)	= 0.03937 × Height	obtained from Equation (12) and actual Height	obtained from Equation (12) with Height = 1244.6 mm

^aShallow only if $V_{oc} < 600$ V ac and the enclosure depth ≤ 203.2 mm (8 in) otherwise the enclosure is “Typical.”

The equivalent enclosure size (*EES*) is determined using the equivalent width and height using Equation (13) as follows:

$$EES = \frac{\text{Height}_1 + \text{Width}_1}{2} \quad (13)$$

where

Height₁ is the equivalent enclosure height
Width₁ is the equivalent enclosure width
EES is the equivalent enclosure size

4.8.4 Determination of enclosure size correction factor (*CF*)

The correction factor (*CF*) for a “Typical Enclosure” is obtained by using Equation (14) as follows:

$$CF = b1 \times EES^2 + b2 \times EES + b3 \quad (14)$$

Use Equation (15) for correction factor for a “Shallow Enclosure” as follows:

$$CF = \frac{1}{b1 \times EES^2 + b2 \times EES + b3} \quad (15)$$

where

b1 to *b3* are the coefficients for Equation (14) and Equation (15) provided in Table 7
CF is the enclosure size correction factor used in Equation (3) through Equation (10)
EES is the equivalent enclosure size used to find the correction factor determined using Equation (13).
For typical box enclosures the minimum value of *EES* is 20

Table 7 provides the coefficients *b1* to *b3* for both typical and shallow enclosure types.

Table 7—Coefficients for Equation (14) and Equation (15)

Box type	E.C.	<i>b1</i>	<i>b2</i>	<i>b3</i>
Typical	VCB	−0.000302	0.03441	0.4325
	VCBB	−0.0002976	0.032	0.479
	HCB	−0.0001923	0.01935	0.6899
Shallow	VCB	0.002222	−0.02556	0.6222
	VCBB	−0.002778	0.1194	−0.2778
	HCB	−0.0005556	0.03722	0.4778

4.9 Determination of I_{arc} , E , and AFB ($600 \text{ V} < V_{\text{oc}} \leq 15\,000 \text{ V}$)

To determine the final arcing current, incident energy, and arc-flash boundary at a specific voltage, first calculate the intermediate values for the three voltage levels of 600 V, 2700 V, and 14 300 V. Then use the interpolation Equation (16) to Equation (24) to determine the final estimated values as follows:

Arcing current (I_{arc})

$$I_{\text{arc}_1} = \frac{I_{\text{arc}_2700} - I_{\text{arc}_600}}{2.1} (V_{\text{oc}} - 2.7) + I_{\text{arc}_2700} \quad (16)$$

$$I_{\text{arc}_2} = \frac{I_{\text{arc}_14300} - I_{\text{arc}_2700}}{11.6} (V_{\text{oc}} - 14.3) + I_{\text{arc}_2700} \quad (17)$$

$$I_{\text{arc}_3} = \frac{I_{\text{arc}_1} (2.7 - V_{\text{oc}})}{2.1} + \frac{I_{\text{arc}_2} (V_{\text{oc}} - 0.6)}{2.1} \quad (18)$$

where

- I_{arc_1} is the first I_{arc} interpolation term between 600 V and 2700 V (kA)
- I_{arc_2} is the second I_{arc} interpolation term used when V_{oc} is greater than 2700 V (kA)
- I_{arc_3} is the third I_{arc} interpolation term used when V_{oc} is less than 2700 V (kA)
- V_{oc} is the open-circuit voltage (system voltage) (kV)

When $0.600 < V_{\text{oc}} \leq 2.7$, the final value of arcing current is given as follows:

$$I_{\text{arc}} = I_{\text{arc}_3}$$

When $V_{\text{oc}} > 2.7$, the final value of arcing current is given as follows:

$$I_{\text{arc}} = I_{\text{arc}_2}$$

The arc duration can be determined using I_{arc} . This time is used to determine the incident energy and arc-flash boundary.

Incident energy (E)

$$E_1 = \frac{E_{2700} - E_{600}}{2.1} (V_{\text{oc}} - 2.7) + E_{2700} \quad (19)$$

$$E_2 = \frac{E_{14300} - E_{2700}}{11.6} (V_{\text{oc}} - 14.3) + E_{2700} \quad (20)$$

$$E_3 = \frac{E_1 (2.7 - V_{\text{oc}})}{2.1} + \frac{E_2 (V_{\text{oc}} - 0.6)}{2.1} \quad (21)$$

where

- E_1 is the first E interpolation term between 600 V and 2700 V (J/cm²)
- E_2 is the second E interpolation term used when V_{oc} is greater than 2700 V (J/cm²)
- E_3 is the third E interpolation term used when V_{oc} is less than 2700 V (J/cm²)

Arc-flash boundary (*AFB*)

$$AFB_1 = \frac{AFB_{2700} - AFB_{600}}{2.1} (V_{oc} - 2.7) + AFB_{2700} \quad (22)$$

$$AFB_2 = \frac{AFB_{14300} - AFB_{2700}}{11.6} (V_{oc} - 14.3) + AFB_{14300} \quad (23)$$

$$AFB_3 = \frac{AFB_1 (2.7 - V_{oc})}{2.1} + \frac{AFB_2 (V_{oc} - 0.6)}{2.1} \quad (24)$$

where

- AFB_1 is the first *AFB* interpolation term between 600 V and 2700 V (mm)
- AFB_2 is the second *AFB* interpolation term used when V_{oc} is greater than 2700 V (mm)
- AFB_3 is the third *AFB* interpolation term used when V_{oc} is less than 2700 V (mm)

When $600 < V_{oc} \leq 2.7$, the final values of incident energy and arc-flash boundary are given as follows:

$$E = E_3$$

$$AFB = AFB_3$$

When $V_{oc} > 2.7$, the final values of incident energy and arc-flash boundary are given as follows:

$$E = E_2$$

$$AFB = AFB_2$$

It is recommended to calculate a second set of arc duration, incident energy, and arc-flash boundary values based on the reduced arcing current I_{arc_min} to account for the arcing current variation effect on the operation of protective devices. The final incident energy or arc-flash boundary is the higher of the two calculated values.

The incident energy (cal/cm²) is obtained by dividing E by 4.184 (1 cal = 4.184 J). See [B.2](#).

4.10 Determination of I_{arc} , E , and AFB ($V_{\text{oc}} \leq 600$ V)

This subclause describes how to determine the final arcing current, incident energy, and arc-flash boundary for a specific open-circuit voltage, $208 \text{ V} \leq V_{\text{oc}} \leq 600 \text{ V}$. First, calculate the arcing current using Equation (25). Using the arcing current, estimate the arc duration and proceed to determine the incident energy and arc-flash boundary.

Arcing current (I_{arc})

The final arcing current can be determined using Equation (25).

$$I_{\text{arc}} = \frac{1}{\sqrt{\left[\frac{0.6}{V_{\text{oc}}}\right]^2 \times \left[\frac{1}{I_{\text{arc_600}}^2} - \left(\frac{0.6^2 - V_{\text{oc}}^2}{0.6^2 \times I_{\text{bf}}^2}\right)\right]}} \quad (25)$$

where

- V_{oc} is the open-circuit voltage (kV)
- I_{bf} is the bolted fault current for three-phase faults (symmetrical rms) (kA)
- I_{arc} is the final rms arcing current at the specified V_{oc} (kA)
- $I_{\text{arc_600}}$ is the rms arcing current at $V_{\text{oc}} = 600 \text{ V}$ found using Equation (1) (kA)

The arc duration can be determined using I_{arc} . This time is used to determine the incident energy and arc-flash boundary.

Incident energy (E)

The incident energy is given as follows:

$$E = E_{\leq 600}$$

where

- $E_{\leq 600}$ is the incident energy for $V_{\text{oc}} \leq 600 \text{ V}$ determined using Equation (6) solved using the arc current determined from Equation (1) and Equation (25) (J/cm^2)
- E is the final incident energy at specified V_{oc} (J/cm^2)

Arc-flash boundary (AFB)

The arc-flash boundary is given as follows:

$$AFB = AFB_{\leq 600}$$

where

- $AFB_{\leq 600}$ is arc-flash boundary for $V_{\text{oc}} \leq 600 \text{ V}$ determined using Equation (10) solved using the arc current determined from Equation (1) and Equation (25) (mm)
- AFB is the final arc-flash boundary at specified V_{oc} (mm)

Calculate a second set of arc duration, incident energy, and arc-flash boundary values based on the reduced arcing current $I_{\text{arc_min}}$ to account for the arcing current variation effect on the operation of protective devices. The final incident energy or arc-flash boundary is the higher of the two calculated values.

4.11 Single-phase systems

This model does not cover single-phase systems. Arc-flash incident energy testing for single-phase systems has not been researched with enough detail to determine a method for estimating the incident energy. Single-phase systems can be analyzed by using the single-phase bolted fault current to determine the single-phase arcing current (using the equations provided in 4.4 and 4.10). The voltage of the single-phase system (line-to-line, line-to-ground, center tap voltage, etc.) can be used to determine the arcing current. The arcing current can then be used to find the protective device opening time and incident energy by using the three-phase equations provided in this guide. The incident energy result is expected to be conservative.

4.12 DC systems

Arc-flash incident energy calculation for dc systems is not part of this model. However, publication references (Ammerman et al. [B1], Das [B16], [B17], Doan [B25], Klement [B62]) provide some guidance for incident energy calculation.

5. Applying the model

The purpose of this clause is to provide an overview of the analysis process required to apply the model. The steps described in this clause may be applied manually, but it may be more convenient to use available short-circuit and protective device coordination programs, which have embedded the steps necessary to apply the model.

Clause 6 provides the following summary of considerations and steps necessary to apply the calculation model:

- An overview of system data collection requirements is provided in 6.2. Accurate data collection is an important part of the study process.
- Calculation of bolted fault current levels, considering system operating modes, is discussed in 6.3 and 6.4.
- Information on equipment-related parameters that are used in the model, such as equipment dimensions, electrode configuration, and working distance, are discussed in 6.5, 6.6, and 6.7.
- Subclauses 6.8 and 6.9 discuss calculation of the arcing current and determination of the arcing duration to be used in the model.
- Subclauses 6.10 and 6.11 address the calculation of the final incident energy and arc-flash boundary. The discussion of specific equipment types in Annex C may also be useful.

To illustrate the model application process, two detailed calculation examples are provided in Annex D.

NOTE—Subclause 4.3 covers the equation application procedure.

6. Analysis process

6.1 General overview

An arc-flash hazard analysis can be performed in association with or as a continuation of a short-circuit study and protective-device coordination study. A complete coordination study may not be required, but the protective device opening time in response to arcing currents must be applied during the analysis process. The process and methodology of calculating short-circuit currents and performing protective-device

coordination is covered in standards such as IEEE Std 551™ (*IEEE Violet Book*™),¹⁰ IEC 60909-0 [B51],¹¹ and IEEE Std 242™ (*IEEE Buff Book*™). The results of the short-circuit study enable calculation of the arcing fault currents at selected locations. Protective device time response to the arcing currents is used to evaluate the time required for the protective devices to interrupt during fault conditions.

Deliverables of the arc-flash hazard analysis calculation are the arc-flash boundary and the arc-flash incident energy at defined working distances from the arcing source at the selected locations in the electrical system. The results of the study document the incident energy analysis and may be used by workers as part of an overall electrical safety risk assessment.

6.2 Step 1: Collect the system and installation data

A significant effort in performing an arc-flash hazard study is the collection of electrical system data. Even for a facility with nominally up-to-date single-line diagrams, time-current curves, and short-circuit model on a computer, the data collection portion of the study may take about half of the effort. Even for new facilities, field verification of the single-line diagrams and protection settings is necessary to verify the integrity of documentation of the power system. Facility workers who are familiar with the electrical system and its safety-related work practices may be able to assist or perform this part of the study. Refer to IEEE Std 1584.1 for further information on system data required for an arc-flash hazard analysis.

While the data required for this study is similar to data collected for typical short-circuit and protective-device coordination studies, it goes further in that all low-voltage distribution and control equipment within the scope of the study up through its sources of supply must be included.

Collect information to perform incident energy calculations on electrical equipment that is likely to require examination, adjustment, servicing, or maintenance while energized. This could include equipment such as low- and medium-voltage switchgear, medium-voltage plug-in connectors, motor starters, motor control centers (MCCs), switchboards, switchracks, panelboards, separately-mounted switches and circuit breakers, ac and dc drives, power distribution units (PDUs), uninterruptible power supplies (UPS), transfer switches, industrial control panels, meter socket enclosures, etc.

The study process begins with a review of available single-line diagrams and electrical equipment site and layout arrangement with people who are familiar with the site. The diagrams should be updated to show the current system configuration.

Electrical system studies should have an up-to-date single-line diagram(s). The single-line diagrams include all alternate feeds.

Follow applicable industry standards for performing short-circuit studies. See [Clause 2](#) for examples of industry standards.

Obtain the available fault current and X/R ratio that represents the source. For transformers, generators, large motors, and switchgear, collect relevant nameplate data such as voltage/voltage ranges or tap settings, ampacity, kilowatt or kilovoltamperes, first cycle (momentary or close and latch) and/or interrupting current rating, impedance or transient/subtransient reactance data. Because information regarding box (enclosure) size and electrode configuration may be needed for more detailed calculations, it may be necessary to take measurements if possible or collect other data such as nameplate information or device catalog numbers that will allow for relevant equipment enclosure dimensions and configurations to be estimated.

Next, collect conductor and cable data along with its installation (routing and support method, in magnetic raceway-steel conduit or non-magnetic raceway-aluminum conduit, etc.) for all electrical circuits between the

¹⁰Information on references can be found in [Clause 2](#).

¹¹The numbers in brackets correspond to those of the bibliography in [Annex A](#).

power source and the distribution and control equipment that is part of the study. This information is needed for the calculation of impedances. See IEEE Std 551 for information on how to perform short-circuit calculations.

Data from instrument transformers (current transformers and voltage transformers) and protective-device data that is part of the study must be collected. The data should be collected from sources such as the nameplate and/or time-current curves. If nameplate data is not accessible, data may be available in specifications or in recent maintenance test reports. In any case, the user should verify data is still up to date by checking with the owner's representative and, if necessary, by checking in the field. In some cases, a field inspection is required to determine the types and ratings of fuses actually installed, as well as the settings of circuit breaker trip units and/or the settings of protective relays. Protective devices that have not been properly maintained may have increased fault clearing time, thereby increasing the incident energy.

Determine which protective device(s) will be used for calculations. The mode of operation, the equipment construction, and the arrangement and characteristics of protective devices (time-overcurrent or otherwise) in an assembly can impact the consideration of which device(s) is selected for calculating the duration of the arc. Engineering judgment by a qualified person with skills and knowledge of the electrical equipment is required for determination of the protective device selected for these calculations. See IEEE Std 1584.1 for further details on the protective device(s) to be considered in arc-flash hazard calculations.

6.3 Step 2: Determine the system modes of operation

An electrical installation may have several modes of operation. It is important to determine the available short-circuit current for the mode(s) of operation that provides both the maximum and minimum available short-circuit currents. See IEEE Std 1584.1 for further details.

A complex power system may have many modes of operation, such as the following:

- a) One or more utility feeders in service
- b) Utility interface substation secondary bus tie circuit breaker open or closed
- c) Unit substation with one or two primary feeders
- d) Unit substation with two transformers with secondary tie opened or closed
- e) MCC with one or two feeders, one or both energized
- f) Generators running in parallel with the utility supply or in standby
- g) Utility system normal switching configured for maximum possible fault megavolt amperes
- h) Utility system normal switching configured for minimum possible fault megavolt amperes
- i) Separately derived sources (generators) – maximum capacity on line
- j) Separately derived sources (generators) – minimum number on line
- k) Shutdown or startup situation with all motors in an off condition – reduced fault contribution

It is necessary to consider the actual modes of operation based on site operating plans, whether in maintenance, normal operation, or under special conditions. Run the incident energy calculations for all modes of operation in the power system to determine the highest incident energy and arc-flash boundary result for each arcing location.

6.4 Step 3: Determine the bolted fault currents

The arc-flash study should be based upon an up-to-date short-circuit study for the facility. The study should take into account both the system data and modes of operation. If an existing study is not available, it will

be necessary to perform one as part of the arc-flash study effort. See IEEE Std 551 and IEEE Std 1584.1 for further details.

Systems containing multiple sources of short-circuit current, such as generators, large motors, or more than one utility supply, can be more accurately modeled with a dynamic simulation method. Methods may include multiple calculations to account for decaying short-circuit current contributions from rotating equipment, and the effect on protective device opening times and resulting incident energy.

Available bolted fault currents should be determined at prospective fault locations based on established standards (see 6.1 for examples of applicable standards). Both larger and smaller available short-circuit currents can result in higher available arc-flash energies and should be considered. Higher fault currents may result in shorter trip times for overcurrent protective devices resulting in a lower incident energy. Higher fault currents without a decrease in the opening time of the overcurrent protective device result in a higher incident energy. Lower fault currents may result in a longer opening time for the overcurrent protective device, thereby increasing the incident energy. If in doubt about the actual fault current, it may be necessary to establish a possible range of fault current levels and calculate the overcurrent protective device tripping times and arc-flash incident energy levels over a range rather than for a specific set of conditions.

6.5 Step 4: Determine typical gap and enclosure size based upon system voltages and classes of equipment

For each piece of equipment that is part of the study, the system voltage and the class of equipment can be used to establish typical gaps between conductors (or bus gaps) as shown in Table 8. It may be difficult to measure the gaps or obtain them from the manufacturer. The gap values provided in Table 8 were derived based on the gaps used in the arc-flash tests. Actual gap measurements from the installed equipment may be used if available, but it may be difficult to establish a single value since the gaps may vary at different locations in the equipment. The typical gaps provided are based on the laboratory test setups and not on actual equipment testing, but they may approximate conductor gaps in actual equipment.

Table 8—Classes of equipment and typical bus gaps

Equipment class	Typical bus gaps (mm)	Enclosure Size (H × W × D)	
		SI units (metric)	Imperial units
15 kV switchgear	152	1143 mm × 762 mm × 762 mm	45 in × 30 in × 30 in
15 kV MCC	152	914.4 mm × 914.4 mm × 914.4 mm	36 in × 36 in × 36 in
5 kV switchgear	104	914.4 mm × 914.4 mm × 914.4 mm	36 in × 36 in × 36 in
5 kV switchgear	104	1143 mm × 762 mm × 762 mm	45 in × 30 in × 30 in
5 kV MCC	104	660.4 mm × 660.4 mm × 660.4 mm	26 in × 26 in × 26 in
Low-voltage switchgear	32	508 mm × 508 mm × 508 mm	20 in × 20 in × 20 in
Shallow low-voltage MCCs and panelboards	25	355.6 mm × 304.8 mm × ≤203.2 mm	14 in × 12 in × ≤8 in
Deep low-voltage MCCs and panelboards	25	355.6 mm × 304.8 mm × >203.2 mm	14 in × 12 in × >8 in
Cable junction box	13	355.6 mm × 304.8 mm × ≤203.2 mm	14 in × 12 in × ≤8 in
		or 355.6 mm × 304.8 mm × >203.2 mm	or 14 in × 12 in × >8 in

Table 8 also provides information on the enclosure sizes used for each voltage class. This information provides the relation between voltage class, gaps, and enclosure sizes. The enclosure sizes were used to derive the enclosure size incident energy correction factor.

6.6 Step 5: Determine the equipment electrode configuration

As part of the calculation process, the equipment conductor and enclosure arrangement that most closely resembles the actual electrode configuration(s) need to be identified. Each type of equipment such as switchgear, panelboards, and motor control centers may contain conductors arranged in similar manner as the test setup electrode configurations presented in Table 9. Locations within a piece of equipment may contain conductor arrangements similar to more than one electrode configuration. As an example, a panelboard may contain both VCB and VCBB electrode configurations. Other types of equipment such as switchgear, disconnect switches, and switchboards may have other electrode configurations such as HCB depending on the bus and conductor arrangement.

Table 9 provides some examples of how equipment conductor arrangements could be classified based on their similarity to the electrode configurations. Depending on the task being performed, and also on the presence (or lack thereof) of removable components, a location can change its electrode configuration classification (e.g., medium-voltage metal-clad circuit-breaker enclosure without the circuit breaker inside the cubicle).

Additional guidance regarding the selection of VCB, VCBB, HCB, HOA, and VOA is available in Annex C and G.2.

Table 9—Correlation between actual equipment and electrode configuration

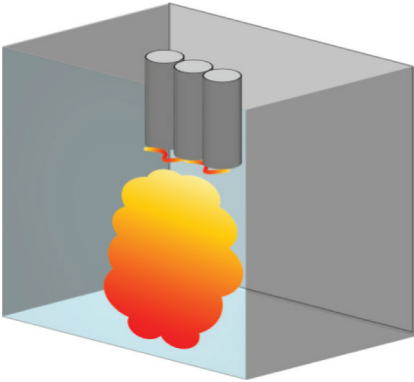
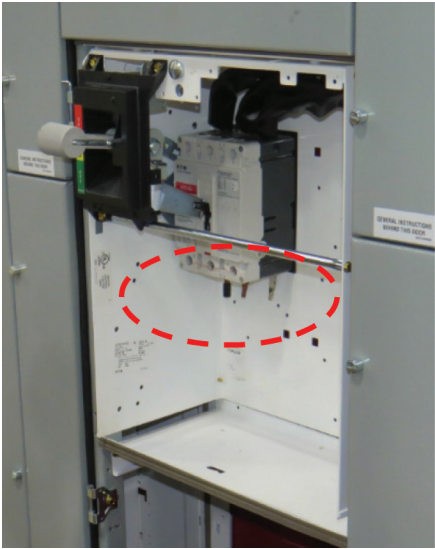
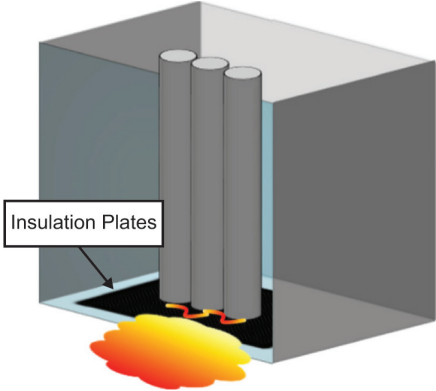
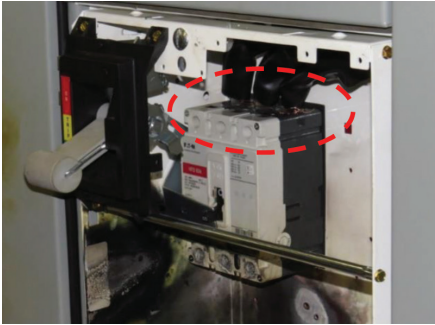
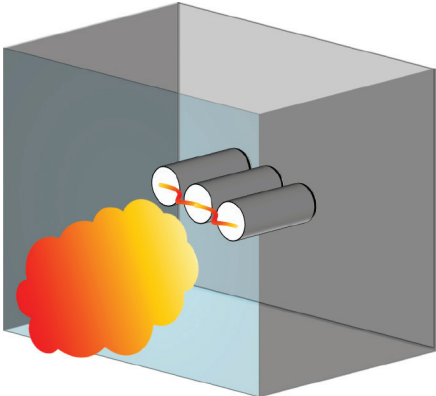
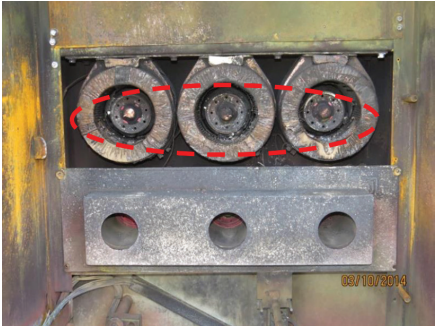
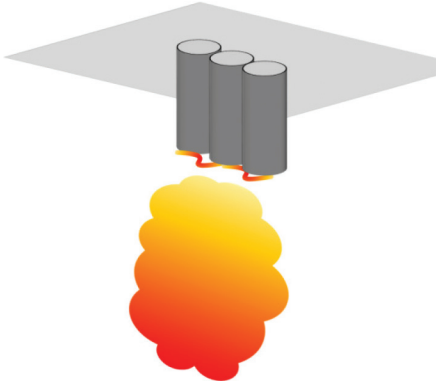
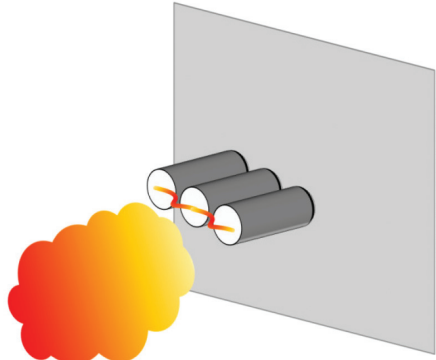
	Electrode configuration in test	Electrode configuration in equipment
VCB		
VCBB		

Table continues

Table 9—Correlation between actual equipment and electrode configuration (continued)

	Electrode configuration in test	Electrode configuration in equipment
HCB	 A 3D diagram showing three cylindrical electrodes protruding from a grey rectangular block. A red and yellow flame-like arc flash is shown at the base of the electrodes.	 A photograph of three circular electrode terminals in a metal enclosure, outlined with red dashed lines. Below them are three circular openings in a metal plate. A date stamp '03/10/2014' is visible in the bottom right corner.
VOA	 A 3D diagram showing three cylindrical electrodes protruding from a grey rectangular block. A red and yellow flame-like arc flash is shown at the base of the electrodes.	 A photograph of high-voltage electrical equipment with multiple red insulators and metal components.
HOA	 A 3D diagram showing three cylindrical electrodes protruding from a grey rectangular block. A red and yellow flame-like arc flash is shown at the base of the electrodes.	 A photograph of three metal terminals labeled X1, X2, and X3 on a green metal enclosure. X1 and X3 are at the top, and X2 is at the bottom.

6.7 Step 6: Determine the working distances

Arc-flash protection is typically based on the incident energy level on the person's head and torso at the working distance and not the incident energy on the hands or arms. Typical working distances can be found in [Table 10](#) based on the class of equipment. The working distance is used in calculating the incident energy. Alternate working distances may be used depending on the task being performed.

Table 10—Classes of equipment and typical working distances

Equipment class	Working distance	
	mm	in
15 kV switchgear	914.4	36
15 kV MCC	914.4	36
5 kV switchgear	914.4	36
5 kV MCC	914.4	36
Low-voltage switchgear	609.6	24
Shallow low-voltage MCCs and panelboards	457.2	18
Deep low-voltage MCCs and panelboards	457.2	18
Cable junction box	457.2	18

6.8 Step 7: Calculation of arcing current

The arcing current depends primarily on the bolted fault current, as well as other factors such as the gap between conductors, electrode or conductor configuration, and system voltage. The available bolted fault current through each protective device is found from the short-circuit study by looking at contributions and impedance of each circuit. Short-circuit current contributions through each circuit connected to the fault location need to be classified as coming from energizing or non-energizing sources or from temporary sources of current such as induction motors. The total arcing current at a given location is calculated based upon the total bolted fault current available at that location. The arcing current distribution among multiple sources is assumed to be the same as the distribution of bolted fault current among the sources.

The arcing current can be calculated by using the equations shown in [Clause 4](#). The calculated arcing current (I_{arc}) is lower than the bolted fault current due to arc impedance. The total arcing current at the point of concern and the portion of that current passing through the upstream protective device(s) must be determined. The portion of the arcing current flowing through the overcurrent protective device determines the duration that is to be used in the incident energy calculation with the total bus arcing current. In the case of locations being energized by multiple feeders, it is necessary to determine the portion of the total arc current passing through each protective device to determine the clearing time for each device.

6.9 Step 8: Determine the arc duration

6.9.1 General

The arc duration is defined as the time it takes the upstream energizing source(s) of arcing current to stop providing current or energy to the arc fault. Typically, the clearing time of overcurrent protective devices depends on the magnitude and/or direction of the arc current passing through their current sensing equipment (current transformers, relays, etc.). When multiple sources are present, the arc duration depends on the time it takes the last protective device to clear the arc current. Under special circumstances, the arc duration is not totally dependent on protective device opening or trip time, but also on the time it takes the stored energy to

be discharged through the arc. Examples of this condition include, but are not limited to, faults near generator terminals on the line side of the generator circuit breaker.

The arc duration is most commonly dependent on the operating time of a time-overcurrent protective device. The operating time of ground-fault relays is not normally considered for the arc duration, as an arcing fault may or may not involve ground. Other types of protective devices with definite operating times such as differential relays, optical arc-flash light-detecting relays, pressure-sensing devices, etc., should be considered to determine their operating time. If protective device configurations are present that affect the operating time of the protective devices, such as zone-selective interlocking schemes, these also need to be considered. All sources of potential delay should be considered, including protective relay operating time, total clearing time of circuit breakers or operating time of contactors, delays introduced by intermediate devices such as lockout or auxiliary relays, delays related to processing times or communication networks, and other factors as appropriate.

For overcurrent protective devices in series, or at locations where more than one type of protective device could clear the arcing fault (e.g., time-overcurrent relay or differential relay), the operating times must be compared to determine which will operate first. During the field survey, up-to-date time-current curves of overcurrent protective devices may have been obtained or developed as part of a coordination study. If not, they should be created to assist in determining the duration of the arc. Commercially available software typically contains extensive overcurrent protective device libraries to aid in the data collection process. When a manufacturer's time-current curve shows a band, or range, the longest time for the calculated arcing current value should be used. These curves will be used to calculate the arc duration based on the average and minimum values of arcing current (I_{arc} and I_{arc_min}) as discussed in [Clause 4](#).

If the total protective device clearing time is longer than two seconds (2 s); consider how long a person is likely to remain in the location of the arc flash. It is likely that a person exposed to an arc flash will move away quickly if it is physically possible, and 2 s usually is a reasonable assumption for the arc duration to determine the incident energy. However, this also depends on the specific task. A worker in a bucket truck, or inside an equipment enclosure, could need more time to move away. Use engineering judgement when applying any maximum arc duration time for incident energy exposure calculations, because there may be circumstances where a person's egress may be blocked.

6.9.2 Fuses

For fuses, information from the manufacturer's time-current curves should be used. These curves may include both melting and total clearing time. If both are available, the total clearing time that represents the worst-case duration should be used. If the curve only consists of the average melt time, 10% of time plus an additional 0.004 s should be added to determine the total clearing time. If the total clearing time at the arcing fault current is less than 0.01 s, then 0.01 s may be used for the time. For current-limiting fuses, if the arcing current is greater than the current-limiting threshold [obtained from the peak let-through (peak let-thru) curves], then use the manufacturer's recommendation on the total clearing time and effective arc current.

A simplified model for some classes of fuses at 600 V and in a VCB configuration is presented in [Annex H](#). Other electrode configurations besides VCB are not considered in [Annex H](#). See [Annex H](#) for a list of the fuse classes, the ratings tested, and the limitations of the application of these models. The manufacturer should be consulted to confirm the appropriateness of these equations.

6.9.3 Low-voltage circuit breakers

For low-voltage circuit breakers with integral trip units, the manufacturer's time-current curves include both the device tripping time and clearing time in most cases. Note that some low-voltage power circuit breakers may be equipped with retrofit trip units. The time-current curves included with the replacement trip unit may, or may not, include the circuit breaker operating time. If the curves show only the trip unit's operating time, a circuit breaker operating time (typically 0.05 s or three cycles) should be added.

A calculation of arc energy with circuit breakers is more accurate when information from the manufacturer's time-current curves is used. However, when they are not available, a conservative method to determine the incident energy based on circuit breakers has been included in [Annex I](#). This method is based on calculated incident energy levels for the VCB configuration only and may be used only if the arc current is in the instantaneous or magnetic trip range.

For current-limiting circuit breakers, if the arcing current is greater than the current-limiting threshold (obtained from the peak let-through curves), then use the manufacturer's recommendation on the total clearing time and effective arc current.

6.9.4 Overcurrent relays and circuit breakers

The manufacturer(s) of the protective relay(s) and circuit breaker(s) should be consulted for detailed information regarding the operating characteristics and time-current curves. For protection schemes using overcurrent protective relays and circuit breakers, the relay time-current curves illustrate the relay operating time. The circuit breaker interrupting time is added to the relay operating time plus any additional time delays such as for lockout-relays, manufacturer's tolerance, and other additional time delay considerations. Circuit-breaker interrupting times can be verified by consulting the manufacturer's literature or the circuit-breaker nameplate data. Interrupting time is the sum of the circuit-breaker opening time and arcing time. See IEEE Std C37.010-2016 and IEEE Std 551-2006 for additional information.

6.10 Step 9: Calculate the incident energy

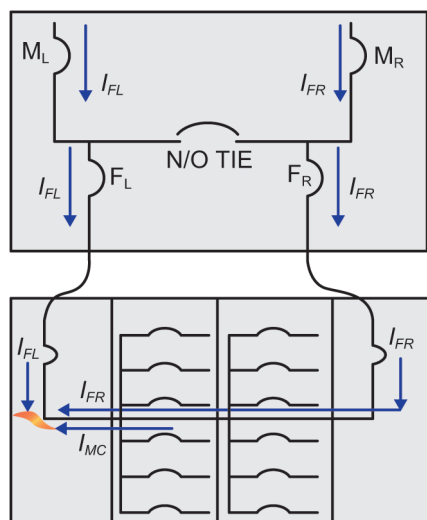
To calculate the incident energy at a specific piece of equipment, the equations in [Clause 4](#) are used. It is important to note that multiple arcing locations can be found within a single piece of equipment as outlined in [6.6](#) and [Annex C](#) and [Annex G](#). Incident energy calculations should be performed at each of the arcing locations that are defined to determine the highest magnitude incident energy or "worst-case" condition.

When a model of the power system is developed, the equipment compartmentalization and fault location need to be considered. The arc fault could occur on the line side, bus side, and load side of protective devices located in different compartments. Refer to IEEE Std 1584.1 for more details on fault location considerations. When evaluating the incident energy at an arcing fault location in the system, the interrupting time of the protective device upstream from the point of the fault must be considered. An integral "main" overcurrent protective device may be considered in the calculation if it is adequately isolated from the bus to prevent escalation to a line-side fault. When the integral main overcurrent protective device is not adequately isolated from the bus, the upstream protective device must be considered as protecting the main and bus.

Systems with motor contribution may require additional considerations for the incident energy calculations. Upstream motor fault current that flows toward the arc-fault location through a feeder may impact the protective device operating time. Downstream motor load contributions also affect the total arcing current and incident energy at the faulted location and need to be accounted for. In other words, the arcing current contribution of motors needs to be considered for its impact on protective device operating time and its effect on the total arcing current and incident energy.

Similar considerations may be needed for systems with multiple sources. Arcing current flows from multiple sources cause protective devices on multiple sources to operate sequentially, causing variation in the arc current flows, arc duration, and incident energy..

[Figure 1](#) shows a dual-source MCC with local mains served from two separate substation feeders with normally open (N/O) tie circuit breaker. A similar concept is applicable for any electrical equipment with multiple sources such as switchgear and switchboards. For the fault on the left side of the MCC main bus, selectivity between the left main (ML) and the left feeder (FL), as well as selectivity between ML and the left MCC main, depends on the current from the left source (IFL). However, the arcing current at the fault consists of current from both main sources (IFL and IFR) and the local motor contribution (IMC).



Source: IEEE Std 1683-2014

Figure 1—Multi-source MCC with motor contribution

The incident energy calculation should consider the change in total arcing current at the fault location caused by the operation of each protective device. The arcing current through each protective device may change based on the removal of other sources of arcing current. The arc energy and incident energy are dependent on the remaining arcing current sources. The total incident energy is then based on the changes to the individual sources of arcing current and their respective durations. The effect of arcing current redistribution after a source of arcing current is removed may also need to be considered. It is possible that the magnitude of the arcing current flowing through a path may change once the impedance of the system changes with each protective device operation. However, the operation and sequence of events presented in this subclause is only an assumption because it is possible that the arc may self-extinguish before the last protective device operation or that it travels to other locations away from its source and develops a different physical behavior.

For each fault current case under consideration, calculate the second incident energy using the minimum arc current and the appropriate arc duration based on the arcing current variation correction factor of 4.5. Choose the higher of the two incident energy values as the calculated incident energy.

6.11 Step 10: Determine the arc-flash boundary for all equipment

To calculate the arc-flash boundary for a given piece of equipment and location, the equations in [Clause 4](#) are used. The arc-flash boundary is the distance from a prospective arc flash where the incident energy is 5.0 J/cm² (1.2 cal/cm²).

6.12 Cautions and disclaimers

As an IEEE guide, this document suggests approaches for conducting an arc-flash hazard analysis but does not contain any mandatory requirements that preclude alternate methods. Following the suggestions in this guide does not guarantee safety, and users should take all reasonable, independent steps necessary to reduce risks from arc-flash events.

Users should be aware that the models in this guide are based upon measured arc-flash incident energy under a specific set of test conditions and on theoretical work. Distances, which are the basis for equations, are based on the measured distance of the test instrument from the arc-flash point source. These models enable users to

calculate the estimated incident energy levels and arc-flash boundary distances. Actual arc-flash exposures may have more or less incident energy than indicated by these models.

This document is intended to provide guidance for the calculation of incident energy and arc-flash boundaries. Once calculated, this information can be used as a basis to develop electrical safety strategies to reduce arc-flash energy exposure.

This information is offered as a tool for conducting an arc-flash hazard analysis. It is intended for use only by qualified persons who are knowledgeable about power system studies, power distribution equipment, and equipment installation practices. It is not intended as a substitute for the engineering judgment and adequate review necessary for such studies.

This guide is based upon testing and analysis of the thermal burn hazard presented by incident energy. Due to the explosive nature of arc-flash incidents, injuries can occur from ensuing molten metal splatters, projectiles, pressure waves, toxic arc by-products, the bright light of the arc, and the loud noise produced. These other effects are not considered in this guide.

This guide is subject to revision as additional knowledge and experience is gained. IEEE, those companies that contributed test data, and those people who worked on the development of this standard make no guarantee of results and assume no obligation or liability whatsoever in connection with this information.

The methodology in this guide assumes that all equipment is installed, operated, and maintained as required by applicable codes, standards, and manufacturers' instructions, and applied in accordance with its ratings. Equipment that is improperly installed or maintained may not operate correctly, possibly increasing the arc-flash incident energy or creating other hazards.

7. Background on the arc-flash hazard

7.1 Early papers

7.1.1 “Arcing fault protection for low-voltage power distribution systems—Nature of the problem” [B59]

This paper identified the potential for personal injury from arcing faults caused by such things as tools contacting bare buses, rodents, dust, insulation failure, or loose connections. The focus was on the nature of arcing faults and the protective equipment and relaying schemes that could be used to extinguish the arc.

7.1.2 “Predicting damage from 277 V single phase to ground arcing faults” [B86]

This paper proposed a method of approximating the degree of burning damage to metal that could be expected from various arcing current values and considerations for coordinating the time and current settings of ground fault protection devices with phase overcurrent protection equipment.

7.1.3 “The other electrical hazard: Electrical arc blast burns” [B67]

The electrical arc-flash hazard was highlighted. The paper described the electrical arc blast as the other electrical hazard. The thermal hazard was described as second-degree burns up to 3.05 m (10 ft) from the arc and third-degree burns up to 1.525 m (5 ft). It also presented theoretical methods of evaluating the open-air arc hazard and gave information on protective measures that should be taken to help avoid or reduce the risk of serious injury.

7.1.4 “The escalating arcing ground-fault phenomenon” [B35]

The possible consequences of arcing ground faults were described in this paper. The phenomena of how low-voltage arcing phase-to-ground faults migrate to three-phase arcs were presented. The observation that the

maximum arcing three-phase fault current is considerably less than the three-phase bolted fault value in 480 V equipment was discussed. The conditions where arcing becomes self-sustaining were described.

7.1.5 “Predicting incident energy to better manage the electric arc hazard on 600 V power distribution systems” [B29]

A method of estimating incident energy on a 600 V, three-phase power distribution system is presented. The effect on incident energy of the arc in a cubic box was considered in developing equations to estimate available bolted fault currents and incident energy at various distances. Benefits of using an estimate of the incident energy in the management of the electrical arc hazard were discussed.

7.1.6 “Report on enclosure internal arcing tests” [B48]

This paper focuses on high-energy arcing faults in enclosures with the compartment door closed. It reports the results of tests in 600 V class MCCs. The need for equipment testing standards in the low-voltage class is identified. Users should identify and provide PPE to personnel working near equipment that cannot contain nor safely vent the arcing hazard.

7.1.7 “Arc and flash burn hazards at various levels of an electrical system” [B55]

This paper presents information from a survey of petrochemical facilities on the PPE used for electrical arc-flash protection. It focuses on the effect of high-energy electrical arcs on humans and presents calculations of distances for curable burn injury at typical industrial/large commercial electrical installations.

7.1.8 “Impact of arc-flash events with outward convective flows on worker protection strategies” [B65]

This paper presents information on additional arc-flash test configurations that increase the convective energy flows outward toward worker. The impact on incident energy predictions is discussed. Further research and modeling method improvements are recommended.

7.2 Additional references

Many other papers have been published on the calculations of arc-flash energy and mitigating the arc-flash hazard, through inherently safer technologies, equipment design improvements, and work practices that reduce the exposure of workers. Reference papers are listed in the bibliography ([Annex A](#)). See [B7], [B13], [B14], [B15], [B18], [B20], [B21], [B24], [B28], [B29], [B31] through [B34], [B36] through [B40], [B44], [B46], [B47], [B48], [B50], [B52], [B54], [B55], [B57] through [B66], [B69], [B70], [B73], [B74], [B76] through [B91], [B96], and [B99].